

Ho Gia Bao (Bao Ho)

Staff Engineer · Singapore

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SUMMARY

Staff Engineer at Tookitaki in Singapore, designing and building backend and data systems for FinCense, the end-to-end platform for anti-money laundering (AML) and fraud prevention used by banks and other financial institutions across Asia Pacific. At Tookitaki since 2018, working on the Transaction Monitoring and Case Manager modules and transaction search on StarRocks, and now building the reliability layer for the platform's data pipelines. In software since 2014 and mostly in Scala since 2016: functional backend services (Scalatra, Cats Effect, Doobie, Circe), Spark jobs, Kafka messaging and the CI/CD around them.

EXPERIENCE

Oct 2018 – now

Tookitaki · Singapore

Staff Engineer Apr 2024 – now

- Designing the reliability layer for FinCense's data pipelines: state guards, idempotent retries, stale-run cleanup, pre-flight validation and metrics.
- Built advanced transaction search on StarRocks end to end, from schema to backend APIs.
- Benchmarked the transaction-monitoring pipeline at 100M+ transactions a day with p90 end-to-end latency under 15 minutes.
- Moved Spark job submission from YARN to Kubernetes, storage from HDFS to S3, HBase data to ScyllaDB, and the Scalatra services to Scalatra 3 and Jakarta EE.
- Fixed memory and latency problems in Case Manager, for example by moving pagination from the JVM into SQL.
- Wrote the architecture and operations runbook for the FinMate AI Copilot for AML investigators: multi-provider LLM serving, streaming, retrieval and tracing.
- Added Prometheus metrics, Grafana dashboards and k6 load tests for the APIs, and built request-reply messaging over Kafka. Mentored three engineers until each could lead their own feature deliveries.

Senior Software Engineer Oct 2018 – Apr 2024

- Reconciliation project: designed the architecture, codebase and database schema from scratch and wrote the boilerplate the team built on. Built user management with LDAP authentication and worked across the Scalatra APIs.
- AML project: built the end-to-end backend APIs from scratch in Scalatra with functional Scala, delivered code with 80% test coverage, and mentored front-end, back-end and DevOps engineers.
- Built a real-time API that sustains 200 transactions per second.
- Cut a MySQL-backed response over 40M+ records from 2 minutes to 10 seconds by reworking the queries and the code path around them.
- Set up CI/CD for the dev, staging and demo environments. Deploy time went from 10 minutes to 30 seconds.
- Maintained an internal data-science platform, worked with data scientists on Spark and MapR jobs, and ran daily stand-ups as Scrum master for teams of 10 and 15.

Scala · Scalatra · Cats Effect · Doobie · Circe · Spark · Flink · Kafka · Kubernetes · AWS · StarRocks · ScyllaDB · MySQL · HBase · Hadoop/MapR · LDAP · Jenkins · Prometheus · Grafana

Apr – Oct 2018

Quoine · Ho Chi Minh City

Data Engineer

- Ran the data platform on Apache NiFi and PostgreSQL and designed the data warehouse.
- Built Tableau dashboards and automated reports for the operations and marketing teams.

Sep 2016 – Jan 2018

EMOBI Pte. Ltd. · Singapore, remote from Ho Chi Minh City

Java/Scala Engineer

- Built a mobile payments platform on Play in Scala and Java: affiliate and publisher management, traffic routing, conversion tracking, scheduled jobs and broadcast messages, and gateway integrations with Mobiltek and Centili.
- Used Akka actors and state machines for distributed workflow processing and modelled data in Cassandra. Set up monitoring (Elasticsearch, Filebeat, Kibana) and CI/CD (Jenkins, GitLab) on Docker, and managed AWS Route 53 and S3.

Jun 2015 – Sep 2016

Mobile Entertainment Corporation · Vietnam

Data Engineer

- Designed and documented a real-time ad-tracking architecture with Nginx, Kafka, Storm, Redis, Elasticsearch, MongoDB and Kibana.
- Built the publisher and advertiser traffic system and a device-fingerprinting service using KNN, SVM and random-forest models.
- Worked on a bank-transfer service (RabbitMQ, Node.js) and push notifications for iOS and Android.

Sep 2014 – Jun 2015

Unicorn · Vietnam

Data Engineer

- Collected Facebook data through the Graph API, ran NLP over posts, and built a model to predict likes on new posts. Used Spark with Scala for a Kaggle competition.

Mar – Sep 2014

Contemi Solutions · Vietnam

Data Engineer Intern

- Set up a Cloudera cluster (Hadoop, Hive, Pig, Sqoop, Spark), built Talend ETL jobs, and trained KNN, SVM and random-forest models in R and Python.

SKILLS

Languages	Scala, Java, Python, SQL, Bash
Backend	Scalatra, Play, http4s, Akka, Cats, Cats Effect, Doobie, Circe, PureConfig, Refined
Data	Apache Spark, Apache Flink, Kafka, RabbitMQ, Hadoop, MapR, Hive, HBase, Storm, NiFi
Databases	MySQL, PostgreSQL, StarRocks, ScyllaDB, Cassandra, MongoDB, Elasticsearch, Redis
Infrastructure	Kubernetes, Docker, AWS, Jenkins, GitHub Actions, GitLab, Nginx, Tomcat, Prometheus, Grafana, Elasticsearch/Kibana
Practices	System design, mentoring, Scrum, writing user stories from client requirements
Spoken	English (professional working proficiency), Vietnamese (native)

EDUCATION

2012 – 2014	Bachelor of Science in Information Technology · VNUHCM University of Science
2008 – 2011	Associate degree in Information Technology · VNUHCM University of Science
2016 – 2017	Functional Programming Principles in Scala; Functional Program Design in Scala; Big Data Analysis with Scala and Spark · EPFL, on Coursera
2015	Tackling the Challenges of Big Data · MIT Professional Education, on edX